

A Target Trial Emulation of ARUBA

A Truveta data request to re-examine treatment versus monitoring in unruptured brain arteriovenous malformations

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Summary

For the past decade, the decision whether to treat or monitor an unruptured brain arteriovenous malformation (AVM) has been influenced by ARUBA, a multicentre randomized trial of 226 patients published in 2014 that concluded monitoring was safer than intervention [1, 3]. That conclusion has remained contentious, criticised for selection bias, the pooling of all treatment types without stratification, and short follow-up, and its uptake has coincided with a rising share of ruptured presentations [5, 8, 13]. A recent effort to correct for these limitations, a 2025 target trial emulation (TTE) using China's nationwide MATCH registry (1,770 patients), reached the opposite conclusion, that intervention improved long-term hemorrhage-free survival [14]. No comparable TTE has been performed in a US, real-world population. We propose to emulate ARUBA in Truveta, representing potentially the largest such emulation to date, and the first to test whether the question can be answered from routine EHR rather than a purpose-built registry. This document specifies the cohort and data required.

Background

A brain arteriovenous malformation (AVM) is a neurovascular lesion in which arteries connect directly to veins through a high-flow tangle of vessels (the nidus), bypassing the normal capillary bed [15]. Left untreated, an unruptured AVM ruptures into a haemorrhagic stroke at roughly 2 to 4 percent per year, and each bleed is fatal in 10 to 30 percent of cases and permanently disables many survivors [15, 16]. Because AVMs usually present in young adults, this risk compounds over decades, making them a leading cause of haemorrhagic stroke before age 50 [16]. For an AVM found before any rupture, clinicians must therefore weigh treating it, by surgery, embolisation, or radiosurgery, each carrying its own stroke risk, against monitoring it and accepting the future rupture risk.

ARUBA remains the only randomized controlled trial ever completed in brain AVMs [1, 3]. It enrolled 226 adults with unruptured AVMs across 39 sites and randomly assigned them to interventional treatment or medical management. The trial was halted early, at a mean of 33 months, after the intervention arm reached the primary endpoint of death or symptomatic stroke more often than the medical arm (30.7 versus 10.1 percent), and concluded that monitoring was safer [1]. Extended follow-up in 2020 upheld that conclusion [2].

The trial drew sustained criticism on several grounds: substantial selection bias and low enrolment, with few high case volume institutions; the pooling of all interventional modalities without stratification; and a follow-up too short to capture a lifelong rupture risk, counting treatment's upfront harms before its long-term benefit could emerge [5, 13, 17, 18]. Its influence on the neuro-interventional field was

immediate and, many argue, harmful. ARUBA shifted practice toward conservative management and lowered interventional treatment rates, yet national data from the years that followed link the post-ARUBA era to a rising proportion of ruptured presentations and worse in-hospital outcomes, suggesting its conclusions may have caused net harm [8]. Given conclusions that are both methodologically fragile and potentially harmful, the treat-versus-monitor question warrants re-examination with modern causal methods, corrected trial design, and larger, more representative populations.

Study Design, Eligibility, and Data

We would conduct a target trial emulation (TTE): a method that first specifies the protocol of the randomized trial one would ideally run, then uses observational data to reproduce it, applying a trial's design and analysis to remove the confounding and immortal-time biases that distort naive observational comparisons [11, 12]. We adopt the operationalisation of the 2025 MATCH emulation [14], and restrict eligibility to the population ARUBA studied. The cohort is defined as follows.

Inclusion criteria

- **Adults.** Age 18 years or older at the index date.
- **Incident AVM.** First recorded diagnosis of cerebral AVM (ICD-10 Q28.2), with no AVM diagnosis during a defined pre-index look-back window.
- **Unruptured.** No intracranial haemorrhage (ICD-10 I60–I62) coincident with or preceding the AVM diagnosis; that is, the AVM was not discovered because of a bleed.
- **Observable at baseline.** Continuous enrolment for a defined pre-index period (for example, 6 to 12 months) to establish incidence and treatment-naïve status.

Exclusion criteria

- **Ruptured presentation.** Haemorrhage (ICD-10 I60–I62) as the mode of, or preceding, AVM diagnosis.
- **Prior AVM treatment.** Any microsurgical resection (CPT 61680–61692), embolisation (CPT 61624 and related), or radiosurgery (CPT 61796–61800; 77371–77373) before the index date.
- **Insufficient baseline data.** Missing essential clinical or imaging information required to define severity and covariates.

Following MATCH [14], each eligible patient is assigned to one of two strategies: interventional treatment initiated within a 6-month grace period of diagnosis, or conservative management (observation and symptom-directed medical therapy); patients not treated within that period are analysed as conservative. Beyond this assignment, we specifically track which type of intervention each patient received (microsurgical resection, embolisation, radiosurgery, or a combination) together with its timing and, where available, whether the AVM was obliterated, so that outcomes can be analysed by modality rather than pooled. This directly addresses ARUBA's most-cited flaw [14]. The primary outcome follows ARUBA, the composite of death or symptomatic stroke, with hemorrhage-free survival at 5 and 10 years reported for comparison [11, 14].

The cohort, treatment strategies, and outcomes can be built largely from standard diagnosis and procedure codes (Table 1); the decisive covariate, AVM severity, must be recovered from radiology-report concepts (Table 2).

Table 1. Cohort, treatment, and outcomes

Data element	Definition / values	Source in Truveta
Cohort entry	Unruptured brain AVM, incident diagnosis	ICD-10 Q28.2
No prior haemorrhage	No I60–I62 before index date	ICD-10 I60–I62 (look-back)
Treatment-naïve	No AVM procedure before index	Procedure codes (look-back)
Demographics	Age ≥18; sex	Structured fields
Intervention arm	Surgery / embolisation / SRS within 6 months, by modality	CPT 61680–61692; 61624 (+S&I); 61796–61800; 77371–77373
Conservative arm	No intervention in grace period	Absence of procedure codes
Outcome: primary	Death or symptomatic stroke	Mortality linkage; ICD-10 I60–I63
Outcome: comparison	Hemorrhage-free survival, 5- and 10-year	ICD-10 I60–I62; mortality linkage
Follow-up	Continuous enrolment per patient; target 5–10 y	Observation window; report median

The standard measure of AVM severity, the Spetzler–Martin grade, is read from angiographic imaging and described in radiology reports rather than captured in diagnosis codes. Because Truveta holds the coded record, these features must be supplied as extracted concepts, and the three Spetzler–Martin components (the first three rows below) are the minimum needed to adjust for confounding by indication. The elements below mirror the angioarchitecture MATCH recorded prospectively [14].

Table 2. AVM severity and angioarchitecture (note-derived concepts)

Data element	Definition / values	Source in Truveta
Nidus size	<3 / 3–6 / >6 cm	Radiology-report concept
Eloquent location	Eloquent vs non-eloquent	Concept
Deep venous drainage	Present / absent	Concept
Spetzler–Martin grade	Derived I–V (from the three above)	Concept if present; else derive
Nidus morphology	Diffuse vs compact (Lawton–Young)	Concept
VALE components	Ventricular involvement; vein aneurysm; deep location; exclusively deep drainage	Concept
AVM location	Lobar / deep / infratentorial / brainstem	Concept
Feeding-artery features	Perforator supply; feeding-artery aneurysm	Concept
Associated aneurysm	Present / absent	Concept / ICD

Covariates for confounding control

- **Include:** age, sex, mode of presentation (seizure [G40], headache, incidental), relevant comorbidities, calendar year of diagnosis, and treating site or region.

What we would like Truveta to confirm

- Which severity components (size, eloquence, deep venous drainage, Spetzler–Martin grade) exist as extracted concepts, and at what coverage and completeness.
- Whether the type of intervention (microsurgery, embolisation, radiosurgery) and, where available, AVM obliteration are captured.
- Cohort counts at each eligibility step.
- Median continuous follow-up available, and whether all-cause mortality linkage is included.

- Cost and lead time to provision the radiology-report concepts in Table 2.

References

1. Mohr JP, et al. Medical management with or without interventional therapy for unruptured brain AVMs (ARUBA): a multicentre, non-blinded, randomised trial. *Lancet* 2014. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4119885/>
2. Mohr JP, et al. ARUBA — final follow-up. *Lancet Neurology* 2020. [https://www.thelancet.com/journals/lanneur/article/PIIS1474-4422\(20\)30181-2/abstract](https://www.thelancet.com/journals/lanneur/article/PIIS1474-4422(20)30181-2/abstract)
3. ARUBA. ClinicalTrials.gov NCT00389181. <https://clinicaltrials.gov/study/NCT00389181>
4. Al-Shahi Salman R, et al. Outcome after conservative management or intervention for unruptured brain AVMs (SIVMS). *JAMA* 2014;311(16):1661–1669. <https://jamanetwork.com/journals/jama/fullarticle/1861801>
5. Treatment and outcomes of ARUBA-eligible patients with unruptured brain AVMs at a single institution. *Neurosurg Focus* 2014;37(3):E8. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4420142/>
6. Interventional outcomes for patients eligible for entry into the ARUBA trial: a systematic review and meta-analysis. *J Neurosurg* 2022;137(1):108. <https://pubmed.ncbi.nlm.nih.gov/34740184/>
7. Dodier P, et al. Outcome of 107 conservatively managed unruptured brain AVMs: a single center's 30-year experience. *J Neurosurg* 2023;139(4):1025–1035. <https://theins.org/view/journals/j-neurosurg/139/4/article-p1025.xml>
8. Impact of the ARUBA trial on trends and outcomes in symptomatic non-ruptured brain AVMs: a national sample analysis. *J Stroke Cerebrovasc Dis* 2022. <https://pubmed.ncbi.nlm.nih.gov/36272182/>
9. Spetzler RF, Martin NA. A proposed grading system for arteriovenous malformations. *J Neurosurg* 1986;65(4):476–483. <https://www.bmc.org/spetzler-martin-grading-scale>
10. Kim H, et al. Validation of the Supplemented Spetzler–Martin (Lawton–Young) grading system in a multicenter cohort of 1009 surgical patients. *Neurosurgery* 2015;76(1):25–33. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4270816/>
11. Hernán MA, Robins JM. Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available. *Am J Epidemiol* 2016;183(8):758–764. <https://academic.oup.com/aje/article/183/8/758/1739860>
12. Hernán MA, Wang W, Leaf DE. Target Trial Emulation: A Framework for Causal Inference From Observational Data. *JAMA* 2022;328(24):2446–2447. <https://jamanetwork.com/journals/jama/article-abstract/2799678>
13. Volovici V, Meling TR. The impact of ARUBA on the clinical practice of unruptured brain arteriovenous malformations: big data, poor evidence and measuring impact on health policy. *Acta Neurochir (Wien)* 2021;163(9):2487–2488. <https://link.springer.com/article/10.1007/s00701-021-04937-6>
14. Han H, Chen Y, Ma L, et al. Interventional Treatment vs Conservative Management of Unruptured Brain AVMs (MATCH registry target trial emulation). *JAMA Netw Open* 2025;8(11):e2543408. <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2841324>
15. Natural history of brain arteriovenous malformations: a systematic review. *Neurosurg Focus* 2014;37(3):E7. <https://theins.org/focus/view/journals/neurosurg-focus/37/3/article-pE7.xml>
16. Intracranial Arteriovenous Malformations. *StatPearls*. <https://www.ncbi.nlm.nih.gov/books/NBK531479/>
17. Bambakidis NC, Cockroft KM, Hirsch JA, et al. The Case Against A Randomized Trial of Unruptured Brain Arteriovenous Malformations: Misinterpretation of a Flawed Study. *Stroke* 2014;45(9):2808–2810. <https://www.ahajournals.org/doi/10.1161/STROKEAHA.114.006519>
18. Viewpoints on the ARUBA Trial. *AJNR Am J Neuroradiol* 2015;36(4):615. <https://www.ajnr.org/content/36/4/615>